Weather Intelligence & Environmental Risk Upgrade Walkthrough

Overview

CropGuardian has evolved from a pure disease detection system into a **Weather-Aware AI Crop Disease Risk & Advisory Platform**. The Google ADK architecture has been expanded to intelligently synthesize visual disease data with real-time geospatial weather conditions.

What's New?

1. **WeatherAgent**: Integrates with Open-Meteo to transparently geocode user location inputs (City, State) or exact coordinates (Lat/Lon). It retrieves high-precision meteorological data (temp, humidity, rain, wind).
2. **EnvironmentalRiskAgent**: Evaluates the retrieved weather against the diagnosed disease. Utilizing deterministic, botanical heuristics, it calculates fungal, bacterial, and heat stress risks to compute an overall spread probability.
3. **Upgraded UI & Advisory**: The Streamlit interface now provides actionable Environmental Intelligence dashboards, and the Gemini-powered Advisory Agent issues precise `Weather-Based Warnings` and `Immediate Actions`.

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1. Updated ADK Workflow Diagram

The workflow forces strict chronological state passing:
`Disease -> Weather -> Severity -> Env Risk -> Advisory`

```
graph TD
    UI[Streamlit UI: Farm Location & Image]
    UI --> ADK[ADK Coordinator Agent]

    subgraph Weather-Aware ADK Workflow
        ADK --> D_Agent[DiseaseAgent]
        D_Agent -- Image --> D_Tool(detect_disease_tool)
```

```
    D_Tool -.-> Legacy_D[DiseaseDetectionAgent]

    D_Tool --> W_Agent[WeatherAgent]
    W_Agent -- Location --> W_Tool(get_weather_tool)
    W_Tool -.-> Legacy_W[WeatherAgent]

    W_Tool --> S_Agent[SeverityAgent]
    S_Agent -- Disease + Image --> S_Tool(analyze_severity_tool)
    S_Tool -.-> Legacy_S[SeverityAgent]

    S_Tool --> R_Agent[EnvironmentalRiskAgent]
    R_Agent -- Disease + Weather --> R_Tool(assess_risk_tool)
    R_Tool -.-> Legacy_R[EnvironmentalRiskAgent]

    R_Tool --> A_Agent[AdvisoryAgent]
    A_Agent -- Disease + Severity + Weather + Risk --> A_Tool(generate_advice_tool)
    A_Tool -.-> Legacy_A[AdvisoryAgent]
  end

  Legacy_A -.-> ADK
  ADK --> |Unified Weather & Disease JSON| UI
```

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2. Updated Project Structure

The agents folder has expanded cleanly:

```
agricare/
    agents/
        adk_workflow/          # ADK Orchestration
        advisory_agent/        # Gemini LLM Integration
        coordinator_agent/      # Legacy Fallback Orchestrator
        disease_detection_agent/ # Visual ML Model
        environmental_risk_agent/# [NEW] Deterministic Risk Scoring
        feedback_agent/        # Model Retraining Feedback
        severity_agent/        # Visual Severity Scoring
        weather_agent/         # [NEW] Open-Meteo Integration
    app/
        pages/
            1_Disease_Detection.py # [MODIFIED] Weather Dashboards
```

```
    main.py
  tests/
    test_weather_agent.py           # [NEW]
    test_environmental_risk_agent.py    # [NEW]
```

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3. Weather Intelligence Walkthrough

When a user visits the Streamlit app:

1. They encounter a new **Farm Location** input block. They can choose to input `City + State + Country` (e.g., "Cherthala, Kerala, India") or raw `Latitude + Longitude`.
2. Behind the scenes, the `WeatherAgent` intercepts this, queries the `geocoding-api.open-meteo.com` to resolve coordinates silently, and then hits `api.open-meteo.com`.
3. The exact resolved location is appended to the payload (`"latitude": 9.6848, "longitude": 76.3356`) to aid in debugging, mapping, and report generation.
4. The UI renders this via 4 crisp metric cards (Temp, Humidity, Precipitation, Wind) directly above the risk assessment.

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4. Environmental Risk Scoring Explanation

> *[!IMPORTANT]*
> *The EnvironmentalRiskAgent is strictly deterministic. It does NOT use LLM reasoning for threshold calculations, making the system highly predictable and scientifically sound.*

How it works:

- **Fungal Risk**: Evaluated strictly on humidity. If humidity > 85%, Fungal Risk is marked as `Very High` (score: 90) because spores rapidly germinate.
- **Bacterial Risk**: Evaluated on a matrix of humidity (>80%) and warm temperatures (25-35°C), or simply rainfall (which causes bacterial splash dispersal).
- **Heat Stress Risk**: Evaluated based on thermal maximums (temp >= 35°C).
- **Spread Probability**: The highest base score (Fungal, Bacterial, or Heat) is taken as the baseline. If an active disease is already detected, a `+10%` penalty is applied to the spread likelihood. If the plant is healthy, a `-20%` reduction is applied.
- **Concept separation**: This provides a distinct metric separating *Severity* ("How badly damaged is the leaf right now?") from *Environmental Risk* ("How likely is the

farm to lose the crop tomorrow?”).